

Enhancing the Digital Inheritance and Embodied Experience of Zen Based on Multimodal Mixed Reality System

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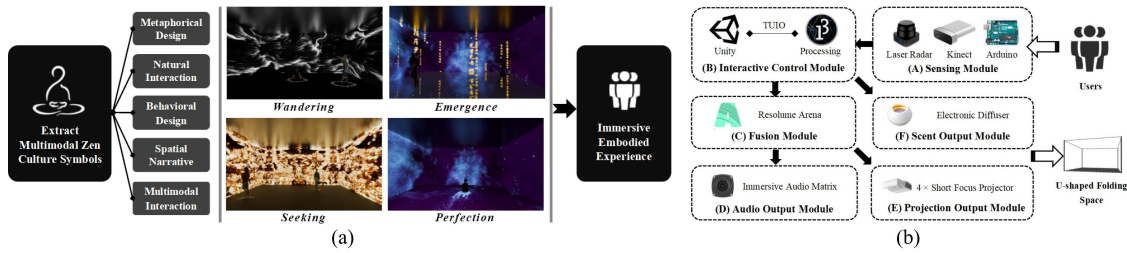


Figure 1: (a) Multimodal scenarios construction process. (b) Technology framework for immersive systems.

ABSTRACT

Zen is an important part of the world's intangible cultural heritage (ICH) but is facing difficulties nowadays. We present an immersive multimodal system for audiences to meditate and experience Zen by integrating AI, natural interaction, and mixed reality (MR). The evaluation ($N = 66$) shows that the system provides the audience with a high-quality sensory experience, enhancing their engagement and empathy. This approach brings new insights into the computer graphic (CG) and ICH community.

KEYWORDS

Zen, Intangible Cultural Heritage, Multimodal Interaction, MR

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1 INTRODUCTION

Inheriting and experiencing traditional cultures is essential to the understanding of human civilization. Among them, Zen holds a significant position in the world. Zen emphasizes mindfulness and contemplation as a spiritual belief and practice, encouraging individuals to focus on inner awareness and achieving a balance between body and mind [Joskovich, 2023]. This is highly meaningful to people currently immersed in high-pressure times.

However, Zen faces challenges in inheritance and accessibility. The main reason is that the form of presentation is too traditional to match the needs of the public, and its concept is too obscure to be approachable [Guo et al., 2023]. Therefore, an attractive system needs to be developed to promote the young generation's participation and to empower the dissemination of Zen.

In recent years, many studies have shown that CG and multimedia technologies have revitalized many tangible cultural heritages [Wu et al., 2023], but there are still few works on ICH. Our study builds on the previous work [Guo et al., 2024] and continues to refine context-building and algorithm development. Facing the dilemma of digitization of ICH (especially philosophical ideas), we have innovatively proposed a novel Immersive MR Space approach, representing a breakthrough from the previous technical solutions like VR and AR. This work provides new insights for the CG & HCI.

2 IMPLEMENTATION

As shown in Figure 1a, the system presents four scenarios, following the sequence of time. We want to make the audiences perceive the flow of time and philosophical questions related to the cycles of life (a metaphor for the central concept of “reincarnation” in Zen).

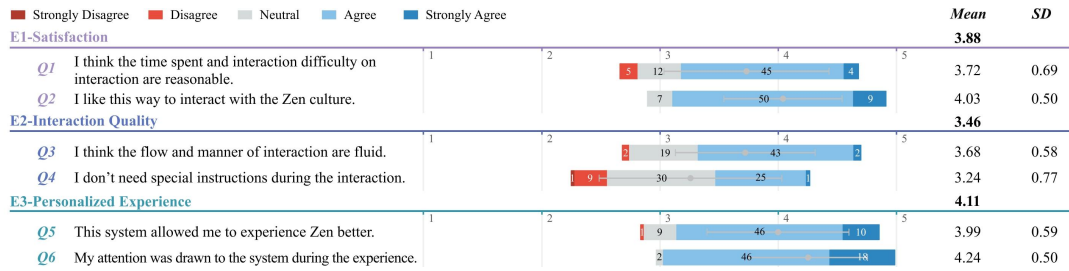


Figure 2: The statistical analysis result of the 5-point Likert questionnaire ($Cronbach \alpha$ & $KMO > 0.7$).

- **Scenario 1: Wandering.** “Wandering” refers not only to the linear symbols but also to the trajectories of all living beings in the world. The dynamic visual symbols, which transcend verbal expressiveness, allow the audience to observe the existence and state of life in the present world (the visual effects are generated in real-time by randomization algorithms). As the audiences enter the space or as the audiences change, the quantity, generation speed, and randomness of the effects will also vary accordingly.
- **Scenario 2: Emergence.** In this scenario, we extract and transform profound and inspiring scriptures into scene, presented in the form of a continuous rain of words. When the audiences read and touch the scriptures, they gradually fade away. This scenario puts the audiences in the rain of falling words, bringing them closer to the ancient scriptures and making the obscure words more accessible.
- **Scenario 3: Exploring.** In this scenario, when the audiences come to the Wall of Buddhas, touch it, the wall peels off (realized by the crushing simulation algorithm), and the back of the Pure Land gradually appears; when the audiences walk, lotus flowers accompany each other, creating the beauty of “step by step, a lotus flower grows”.
- **Scenario 4: Perfection.** Perfection is the final scenario, guiding the audience’s vision back to the universe. In immersive setting, the audiences have the freedom to wander around or engage in meditation, allowing them to experience “emptiness”. The scent of sandalwood fills the air, enhancing the tranquil atmosphere. When the audience has folded their hands and is in a calm mood (realized by motion capture, and physiological computing algorithms), the Seal of Wishes moves to the ground of the audience, which means “fulfilling and realizing their aspirations”.

Figure 1b shows the technical framework and workflow of the system. After the audience enters the interactive area, (a) Sensing Module acquires the number, position, action, and physiological data of the audience through Radar, Kinect, and Arduino, then transmits it to (b) Interactive Content Control System. (b) generates interactive effects according to algorithms, then transmits the data through the Syphon/Spout protocol to (c) Fusion Module. (c) fuses, divides, corrects, and distributes the scenes according to the size and shape of the actual exhibition space, and transmits them to (d) Audio Output Module and (e) Projection Output Module. (d) and (e) output digital or physical content in the exhibition space.

3 PRELIMINARY EVALUATION

We conducted a pilot study to measure 66 audiences’ (Aged 19–72, $M = 30.4$, $SD = 11.1$, *Females* = 40) satisfaction, interaction quality, and personalization by mixed methods based on [Guo et al., 2023; Jin et al., 2021]. The statistical results are shown in Figure 2.

Results show that *Digital Pure Land* is appreciated by audiences of different cultural backgrounds and ages regarding satisfaction and personalized experience. Many audiences said, “My heart was cleansed and it made me reflect on my inner world”, “Meditating in the void, I gradually forgot myself and my surroundings”. These results indicate that the immersive system helps their understanding and awareness of the context, bridging the gap to ancient philosophy.

It is worth mentioning that some young audiences also took photos or videos and shared them with feelings on social media, promoting the dissemination of Zen. However, there is still room for optimization in interaction quality ($M = 3.46$), like guiding the process, controlling the number of audiences, etc.

4 DISCUSSION AND CONCLUSION

Taking Zen as a case, we designed and developed an immersive MR system *Digital Pure Land*. Preliminary evaluation shows that our approach had a significant positive effect on the UX and Learning. In the future, we will continue to explore the digitization of ICH.

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